Calculating Variance of Weekly Step Counts

Given data: [4500, 7000, 5000, 8000, 4000, 9000, 6000]

We will calculate the **population variance**, treating these 7 days as the full week.

Step 1: Find the mean (average)

Mean = Sum of all values / Number of values

Sum = 4500 + 7000 + 5000 + 8000 + 4000 + 9000 + 6000 = 43,500

Mean = 43,500 / 7 = **6,214.29 steps** (approximately)

Step 2: Find each deviation and square it

Day's steps (x)	Deviation (x – mean)	Squared deviation (x – mean)²
4,500	–1,714.29	2,938,775.51
7,000	785.71	617,346.94
5,000	–1,214.29	1,474,489.80
8,000	1,785.71	3,188,775.51
4,000	–2,214.29	4,903,061.22
9,000	2,785.71	7,760,204.08
6,000	–214.29	45,918.37
Total	**0 (approximately)**	**20,928,571.43**

Step 3: Calculate variance

Population variance = Sum of squared deviations / Number of values

Variance = 20,928,571.43 / 7 = **2,989,795.92 steps²**

Final answer: **Population variance ≈ 2,989,795.92 steps²**

Note: Small differences may occur because the mean and squared deviations are rounded. If the question asks for sample variance, divide by n – 1 = 6 instead of 7.